Exercise 2.4 If the step-size parameters, α_n, are not constant, then the estimate Q_n is a weighted average of previously received rewards with a weighting different from that given by (2.6). What is the weighting on each prior reward for the general case, analogous to (2.6), in terms of the sequence of step-size parameters? □

$$Q_{n+1}(a) = Q_n(a) + \alpha_n(a)\left[R_n - Q_n(a)\right] =$$

$$= Q_n(a) + \alpha_n(a) R_n - \alpha_n(a) Q_n(a) =$$

$$= \boxed{\alpha_n(a) R_n + \left(1 - \alpha_n(a)\right) Q_n(a)} =$$

$$= \alpha_n(a) R_n + \left(1 - \alpha_n(a)\right)\left[\alpha_{n-1}(a) R_{n-1} + \left(1 - \alpha_{n-1}(n)\right) Q_{n-1}\right] =$$

$$= \alpha_n(a) R_n + \left(1 - \alpha_n(a)\right)\alpha_{n-1}(a) R_{n-1} + \left(1 - \alpha_n(a)\right)\left(1 - \alpha_{n-1}(a)\right) Q_{n-1} =$$

$$= \alpha_n(a) R_n + \left(1 - \alpha_n(a)\right)\alpha_{n-1}(a) R_{n-1} + \left(1 - \alpha_n(a)\right)\left(1 - \alpha_{n-1}(a)\right) Q_{n-1} +$$

$$+ \dots \alpha_1(a) \prod_{i=2}^{n} \left(1 - \alpha_i(a)\right) R_1 + \prod_{i=1}^{n} \left(1 - \alpha_i(a)\right) Q_1 =$$

$$= \prod_{i=1}^{n} \left(1 - \alpha_i(a)\right) Q_1 + \sum_{i=1}^{n} \left(\alpha_i(a) \prod_{j=i+1}^{n} \left(1 - \alpha_j(a)\right) R_i\right)$$